

Multiple Unit Synchronization Not Active When Requested

Symptoms

- Multiple Unit Synchronization Active when **not** Requested.
- ECM is **not** communicating with customer interface box panel.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

The multiple unit synchronization switch 2 and 1 signal circuits are an option for the CIB panel.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box.	
	STEP 1A. Check the DCU410 unit display for faults.	
	STEP 1A-1. Check the DCU410 unit power supply wire for voltage +24-VDC.	
	STEP 1B. Check the multiple unit synchronization circuits for an open.	
	STEP 1C. Check the multiple unit synchronization circuits for a wire-to-wire short.	
	STEP 1D. Check the multiple unit synchronization circuits for a short to ground.	
STEP 2.	Check the OEM wiring harness.	

STEPS		SPECIFICATIONS
	STEP 2A. Check the multiple unit synchronization circuits for an open.	
	STEP 2B. Check the multiple unit synchronization circuits for a wire-to-wire short.	
	STEP 2C. Check the multiple unit synchronization circuits for a short to ground.	

STEP 1. Check the customer interface box.

STEP 1A. Check the DCU410 unit display for faults.

Conditions :

- Locate the DCU410 unit display.

Action	Specification/Repair	Next Step
Check the DCU410 unit display for indication of faults.	DCU410 unit indicates fault(s)?	1B
	YES	
	DCU410 unit indicates fault(s)?	1A-1
	NO	

STEP 1A-1. Check the DCU410 unit power supply wire for voltage +24-VDC.

Conditions :

- Open the customer interface box.

Action	Specification/Repair	Next Step
--------	----------------------	-----------

Check the voltage at the battery 1 voltage (switched power) at the DCU410 unit. - Place one test on the battery 1 voltage (switched power) wire at the DCU410 unit. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than +24-VDC? YES Repair : Check the batteries. Refer to OEM service manual.	Repair complete
	Less than +24-VDC? NO	1B

STEP 1B. Check the multiple unit synchronization circuits for an open.		
Conditions : - Open the customer interface box. - Disconnect the multiple unit synchronization switch 2 signal wire from the X4 and C2 connectors. - Disconnect the multiple unit synchronization switch 1 signal wire from the X4 and C2 connectors.		
Action	Specification/Repair	Next Step

Check the multiple unit synchronization switch signal wires for an open. - Place one test lead on the multiple unit synchronization switch 1 signal wire at the X4 connector. - Place the other test lead on the multiple unit synchronization switch 1 signal wire at the C2 connector. - Place one test lead on the multiple unit synchronization switch 2 signal wire at the X4 connector. - Place the other test lead on the multiple unit synchronization switch 2 signal wire at the C2 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1C
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete

STEP 1C. Check the multiple unit synchronization circuits for a wire-to-wire short.

Conditions : - Open the customer interface box. - Disconnect the multiple unit synchronization signal wires at the X4 connection.		
Action	Specification/Repair	Next Step

Check the multiple unit synchronization switch signal wires for a wire-to-wire short. - Place one test lead on the multiple unit synchronization switch 1 signal wire at the X4 connection. - Place the other test lead on all other wires at the X4 connection. - Place one test lead on the multiple unit synchronization switch 2 signal wire at the X4 connection. - Place the other test lead on all other wires at the X4 connection. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1D
	Less than 10 ohms? NO Repair : Replace the X4 connection. Contact a Cummins® Authorized Repair Location.	Repair complete

STEP 1D. Check the multiple unit synchronization circuits for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the CLU connector.

Action	Specification/Repair	Next Step

Check the multiple unit synchronization switch signal wires for a short to ground. - Place one test lead on the multiple unit synchronization switch 1 signal wire at the X4 connection. - Place the other test lead on panel ground. - Place one test lead on the multiple unit synchronization switch 2 signal wire at the X4 connection. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the X4 connection. Contact a Cummins® Authorized Repair Location.	Repair complete
	Less than 10 ohms? NO	2A

STEP 2. Check the OEM wiring harness.

STEP 2A. Check the multiple unit synchronization circuits for an open.

Conditions :

- Locate the OEM wiring harness.
- Disconnect the multiple unit synchronization switch 1 signal wire from the C2 and 50-pin ECM connector.
- Disconnect the multiple unit synchronization switch 2 signal wire from the C2 and 50-pin ECM connector.

Action	Specification/Repair	Next Step

Check the multiple unit synchronization switch signal wires for an open. - Place one test lead on the multiple unit synchronization switch 1 signal wire at the C2 connector on panel. - Place the other test lead on the multiple unit synchronization switch 1 signal wire at the 50-pin ECM connector. - Place one test lead on the multiple unit synchronization switch 2 signal wire at the C2 connector on panel. - Place the other test lead on the multiple unit synchronization switch 2 signal wire at the 50-pin ECM connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	2B
	Less than 10 ohms? NO Repair : Replace the wire. Refer to the OEM service manual.	Repair complete

STEP 2B. Check the multiple unit synchronization circuits for a wire-to-wire short.

Conditions :

- Open the customer interface box.

Action	Specification/Repair	Next Step

Check the multiple unit synchronization switch signal wires for a wire-to-wire short. - Place one test lead on the multiple unit synchronization switch 1 signal wire at the C2 connector. - Place the other test lead on all other wires at the C2 connector. - Place one test lead on the multiple unit synchronization switch 2 signal wire at the C2 connector. - Place the other test lead on all other wires at the C2 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace connector. Refer to Procedure 019-208 (Deutsch®; HD20 and HD30 Connector Series) in Section 19 in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850 Modular Common Rail System, Bulletin 4021493 or the Troubleshooting and Repair Manual, Electronic Control System, QSK38, QSK50, and QSK60 (CM850 Modular Common Rail System), Bulletin 4021533. (/qs3/pubsys2/xml/en/procedures/99/99-019-208.html) Less than 10 ohms? NO	Repair complete 2C
--	--	--

STEP 2C. Check the multiple unit synchronization circuits for a short to ground.

Conditions : Open the customer interface box.		
Action	Specification/Repair	Next Step

Check the multiple unit synchronization switch signal wires for a short to ground. • Place one test lead on the multiple unit synchronization switch 1 signal wire at the C2 connector. • Place the other test lead on panel ground. • Place one test lead on the multiple unit synchronization switch 2 signal wire at the C2 connector. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace connector. Refer to Procedure 019-208 (Deutsch®; HD20 and HD30 Connector Series) in Section 19 in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850 Modular Common Rail System, Bulletin 4021493 or the Troubleshooting and Repair Manual, Electronic Control System, QSK38, QSK50, and QSK60 (CM850 Modular Common Rail System), Bulletin 4021533. (/qs3/pubsys2/xml/en/procedures/99/99-019-208.html)	Repair complete
	Less than 10 ohms? NO	Contact a Cummins® Authorized Repair Location.